How a saved report becomes an alert

The server does the checking. The app only draws the result.

1

Save

The patient confirms the values. They are stored.

2

Trigger

The database calls the FastAPI service.

3

Load

FastAPI reads the full history of each test.

4

Check

Four rules run on every test in the report.

5

Alert

A finding becomes a push and an in-app alert.

Clean the data before any rule runs

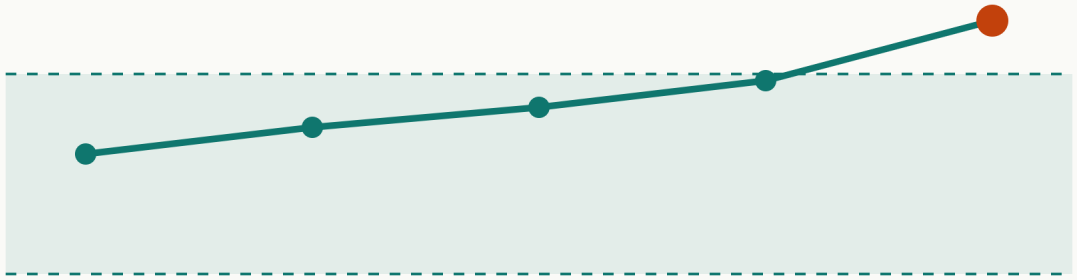
Rules only work on a tidy series, so the server prepares it first.

Problem	Raw example	After cleaning
Test names	“Hb”, “HGB”, “Haemoglobin”	One test: Hemoglobin
Units	Glucose 5.9 mmol/L	About 106 mg/dL
Dates	3 Jan, 10 Jan, 14 Mar	Day 0, day 7, day 70
Short history	Only 2 reports so far	Only the range check runs

Real day gaps matter: a six-month gap between reports is not treated like a one-week gap.

Rules 1 and 2: range and rises

RULE 1 · OUT OF RANGE



Glucose in mg/dL: 88, 92, 95, 99, 108. The shaded band is the 70–100 range printed on the report. The latest value sits 8 above it, so the app flags it.

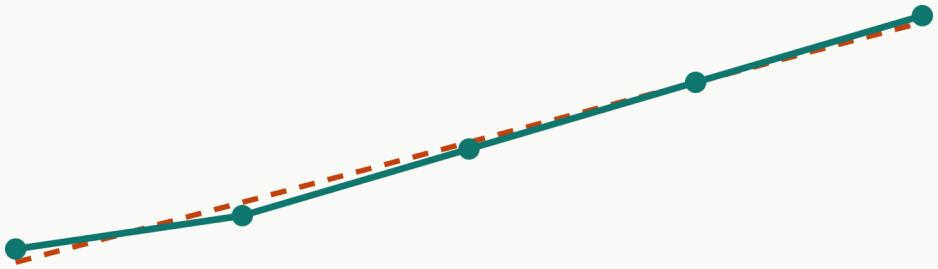
RULE 2 · CONSECUTIVE RISES



Glucose in mg/dL: 92, 96, 101, 107. Each step is more than 2% above the last, so three rises in a row are flagged. Wobbles under 2% are ignored as lab noise.

Rules 3 and 4: slope and jumps

RULE 3 · STEEP SLOPE



HbA1c in %: 5.4, 5.5, 5.7, 5.9, 6.1 over one year. A straight line fitted through them (dashed) rises about 1% per month and fits closely. The example limit for this test is 0.8%, so it is flagged.

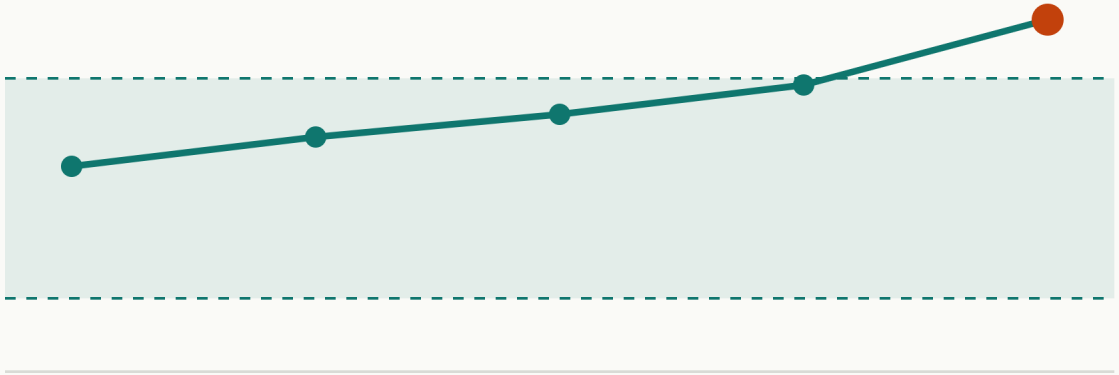
RULE 4 · SUDDEN JUMP



Glucose in mg/dL: 95, 97, 96, 98, then 126 five days later. That is +29% from the previous report, above the example 25% limit. The same change over a year would count as less urgent.

What the patient finally sees

Glucose (mg/dL), last five reports



Possible trend

Glucose: possible trend

Out of range, and up four reports in a row (+23%).
Latest: 108 mg/dL, range 70–100.

Not a diagnosis. Consider consulting a doctor.

View graph | Share with a doctor

No repeat alert unless the trend gets worse.

One alert per test, listing every rule that fired.

Thresholds are set per test by the medical advisor.